

Visualization Design Canvas

Template

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Example Project

Gapminder — From Data to Visual Language

Good visualizations are designed, not generated.

DN-001 — Dataset Card

Dataset

Title

Gapminder

Source

Gapminder Foundation

<https://www.gapminder.org/>

Original author

Gapminder Foundation

Description

The Gapminder dataset contains demographic, economic and health indicators collected for countries around the world over multiple years.

Each observation represents one country in one year.

The dataset combines several variables that allow exploration of relationships between wealth, health and population over time.

Observations

One row = one country in one year.

Variables

Variable	Type	Description
country	categorical	Country name
continent	categorical	Continent
year	temporal	Observation year
lifeExp	continuous	Life expectancy
pop	continuous	Population
gdpPercap	continuous	GDP per capita

Data quality

Missing values

Very few.

Outliers

Countries with extremely large populations.

Countries with very high GDP per capita.

Potential bias

Countries differ greatly in population size.

Economic indicators are estimates.

Initial observations

Health and wealth seem related.

Population varies by several orders of magnitude.

Time appears to play an important role.

DN-002 — Audience & Goal Card

Audience

Students interested in data visualization.

Basic statistical knowledge.

Little experience with visualization design.

Context

Presentation during a workshop.

Main objective

Explain the relationship between economic development and health.

Key message

Countries become healthier as wealth increases, but development is not uniform.

Main question

How do health, wealth and population change over time?

Success criteria

The audience should immediately recognize:

- relationship between GDP and life expectancy
- differences between continents
- population size
- temporal evolution

DN-003 — Data Understanding Card

Variables of interest

Variable	Type	Importance
GDP per capita	continuous	High
Life expectancy	continuous	High
Population	continuous	High
Continent	categorical	High
Year	temporal	High

Relationships

GDP vs. Life expectancy

Population vs/ Country importance

Continent vs. Regional differences

Year vs. Development

Interesting questions

How does wealth influence life expectancy?

Do continents evolve differently?

Which countries improve the fastest?

What happens through time?

Hypothesis

Countries generally become healthier and wealthier over time.

DN-004 — Visualization Mapping Card

Data → Visual Encoding

Data	Encoding	Why
GDP per capita	X position	Accurate quantitative comparison
Life expectancy	Y position	Accurate quantitative comparison
Population	Bubble size	Magnitude
Continent	Color	Category recognition
Year	Animation	Temporal evolution

Visual hierarchy

First

Overall relationship between wealth and health.

Second

Large countries.

Third

Continental patterns.

Accessibility

- Colorblind-friendly palette
- High contrast
- Minimal clutter
- Simple labels

DN-005 — Representation Card

Candidate representations

Option A

Scatter plot

Advantages

- relationship visible
- simple
- accurate

Limitations

- no population
- no time

Option B

Bubble chart

Advantages

- adds population

Limitations

- bubble area harder to compare

Option C

Animated bubble chart

Advantages

- shows temporal evolution
- engaging
- reveals trajectories

Limitations

- harder to compare specific years

Decision

Animated Bubble Chart

Reason:

It communicates five variables simultaneously while maintaining readability.

DN-006 — Implementation Card

Tool

R

ggplot2

gganimate

Why?

Grammar of Graphics provides a declarative way to map data onto visual attributes.

Animation naturally represents temporal change.

Workflow

1. Import data
2. Explore variables
3. Design mapping
4. Prototype
5. Refine
6. Review

DN-007 — Review Card

Message

- Clear
- Focused
- Honest

Audience

- Appropriate
- Easy to understand

Design

- Effective color
- Good visual hierarchy
- Appropriate labels
- Limited chartjunk

Reflection

Strengths

- multiple variables
- intuitive mapping
- memorable

Weaknesses

- animation prevents precise comparison
- bubble area is less accurate than position

Possible improvements

- small multiples
- interactive controls
- annotations

Final Design Summary

Goal

Explain global development.

Audience

Students learning visualization.

Question

How are wealth, health and population related over time?

Representation

Animated bubble chart.

Mapping

GDP -> X

Life expectancy -> Y

Population -> Size

Continent -> Color

Year -> Animation

Implementation

Possible `ggplot2` + `gganimate`

“The visualization was not chosen because it looks attractive.

It was chosen because every visual attribute supports the communication goal.”

Implementation details

AI prompt i implementation code.

DN-008 — Implementation Prompt *Exemplary prompt for AI-assisted implementation:*

Create an R implementation of an animated Gapminder bubble chart.

Use the `gapminder` dataset.

The visualization should show:

- GDP per capita on the X axis,
- life expectancy on the Y axis,
- population as bubble size,
- continent as color,
- year as animation frame.

Use `ggplot2` and `gganimate`.

Use a log scale for GDP per capita.

Make the chart readable, clean and suitable for a presentation.

Add clear title, subtitle, axis labels and legend labels.

Export the animation as a GIF file.

Remarks:

- A prompt should describe design decisions, not code.
- The code is simply the implementation of those decisions in R.

DN-009 — R Implementation

N-010 — Static Alternative

Final Thoughts

Today, the most valuable skill is no longer writing visualization code. Instead it is making and communicating sound design decisions. The code is merely an implementation; the design is where visualization begins.

The key competency of the future is not mastering programming syntax or visualization tools, but making sound design decisions and expressing visualization requirements with clarity.